

Hydrogen Production via Steam Reforming of Glycerol Using a Gibbs Reactor in Aspen HYSYS

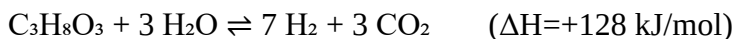
Mr. Ivica MARJANOVIC

UPMC – Université Pierre et Marie Curie, Paris – CAPGEMINI, Paris

Background & Description:

This project investigates the simulation and evaluation of hydrogen production from **steam reforming of glycerol**, using **Aspen HYSYS** as the process modeling tool. Glycerol is a major by product of biodiesel production, and its direct reuse is often economically unfeasible due to impurities. Converting glycerol into hydrogen presents a sustainable pathway for clean fuel generation.

The process is governed by the main reforming reaction:



The simulation flowsheet includes the following key steps:

- **Feed Preparation:** Pure glycerol and water are mixed at ambient conditions (25 °C, 1 atm) in a 3:1 molar ratio.
- **Heating:** The mixture is heated up to 600–658 °C using a heat exchanger.
- **Reaction:** The hot mixture enters a **Gibbs Reactor**, where equilibrium-based reaction calculations are performed. This type of reactor does not require manual reaction input; instead, it computes the most thermodynamically favorable product distribution by minimizing the Gibbs free energy.
- **Cooling:** The reactor outlet stream is cooled back to 25 °C.
- **Separation:** A **Separator** is used to split the outlet stream into a hydrogen-rich vapor phase and a water-rich liquid phase.
- **Recycling:** The separated water is recycled back into the feed to improve efficiency and reduce freshwater consumption.

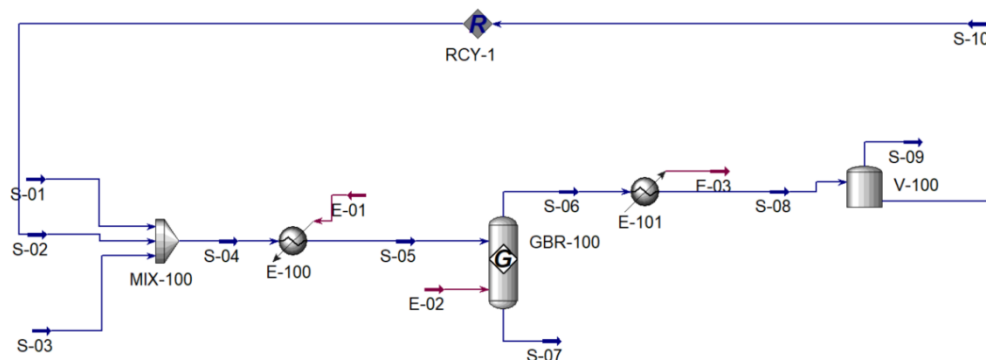
This flowsheet was implemented using the **Peng-Robinson equation of state**, which is suitable for high-temperature gas-phase systems.

Simulation results indicate a hydrogen production rate of approximately **5.4 mol H₂ per mol of glycerol**, with hydrogen representing up to **62–66% molar fraction** in the gas phase, under optimal conditions (658 °C, 1 atm). These results align closely with published literature, validating the effectiveness of the Gibbs reactor model for this type of thermodynamic simulation.

This project highlights Aspen HYSYS's capability to model complex equilibrium-based systems using built-in tools such as the Gibbs reactor and recycle streams, offering a practical and efficient way to simulate hydrogen production from renewable feedstocks.

Thermodynamic package: The Peng-Robinson property method was selected, as it effectively models gas-phase equilibria and non-ideal behavior at elevated temperatures and pressures, making it appropriate for the steam reforming system.

Flowsheet



Results:

Master Property Tableau

Object	S-02	S-04	S-05	S-07	S-06	S-08	S-09	S-10
T (°C)	25.00	24.08	600	600	600	25.00	25.00	25.00
P (bar)	1.013	1.013	1.013	1.013	1.013	1.013	1.013	1.013
Molar Flow (kgmole/h)	17.33	21.33	21.33	0	27.30	27.30	9.891	17.41
Mole Fract (Carbon)	0	0	0	0	0	0	0	0
Mole Fract (CO)	0	0	0	0	0.0131	0.0131	0.0360	0
Mole Fract (CO2)	0.0001	0.0001	0.0001	0	0.0964	0.0964	0.2658	0.0001
Mole Fract (Glycerol)	0	0.0469	0.0469	0	0	0	0	0
Mole Fract (H2O)	0.9999	0.9530	0.9530	0	0.6489	0.6489	0.0310	0.9999
Mole Fract (Hydroge n)	0	0	0	0	0.2412	0.2412	0.6657	0
Mole Fract (Methane)	0	0	0	0	0.0005	0.0005	0.0015	0

Nb : Insignificant quantities (<0) of components in the product streams can be considered negligible, effectively approximating them to zero without impacting the overall process results or conclusions (see flowsheet).

References: Haider, M. A., & Chaturvedi, N. D. (2023). An energy-efficient and cleaner production of hydrogen by steam reforming of glycerol using Aspen Plus. *International Journal of Hydrogen Energy*